

TIANYUAN (TIANA) WANG

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SUMMARY

Applied economics Ph.D. candidate and former IMF Research Analyst with hands-on experience in data science. Uses causal inference and large-scale transaction data to study how pricing and policy changes affect consumers and markets. Builds reproducible analytical workflows and decision-support tools for economists and industry clients.

EDUCATION

University of Georgia <i>Ph.D. in Agricultural and Applied Economics; concurrent M.S. in Statistics</i>	2023–2027 (expected)
Johns Hopkins University <i>M.S. in Government Analytics (Data Analytics and Policy)</i>	2018–2020
Hong Kong Baptist University <i>B.B.A. in Applied Economics; minor in Finance</i>	2014–2018

SELECTED RESEARCH

The Food-Price Effect of H-2A Wage Mandates Presented at AAEA 2026
With Michael Adjemian and Genti Kostandini; Job market paper

- Study how the Adverse Effect Wage Rate (AEWR) affects farm labor and how wage-floor increases pass through to U.S. food prices; preliminary retail-price estimates indicate a positive price response.
- Apply continuous-treatment LP-DiD to QCEW employment and payroll using 2002 migrant-labor farm exposure and AEWR-premium shocks; preliminary estimates show higher employment, with no clear payroll increase.
- Design midstream analyses of farm-gate prices and import quantities by products, comparing crop labor intensity and AEWR exposure across producing regions to study price transmission and import adjustment.
- Constructed county-by-category monthly price indices from Nielsen retail scanner data, using SLURM arrays and integrity checks to process an 8 TB raw input tree on high-performance computing cluster.

Food Price Inflation in the United States Presented at SAEA 2026
With Michael Adjemian and Qingxiao Li; NIFA-funded

- Replicated the national PCE supply/demand inflation decomposition and implemented rolling bivariate VARs with residual-sign classification to extend the analysis to county-level price and quantity data.
- Built a restartable county-level pipeline aligning price, quantity and expenditure data; validated the implementation on simulated data ahead of full-sample estimation.

PROFESSIONAL EXPERIENCE

International Monetary Fund | Research Analyst Dec 2020–Jul 2023
Washington, DC | African Department, Southern Africa Division | Top 10 Outstanding Performance

- Produced data-driven analyses with story-telling figures for IMF Article IV Staff Reports. [Excel, SQL]
- Maintained the macroeconomic framework in four sectors (real, fiscal, external, financial) and validated WEO submissions, tracing discrepancies to formula errors, missing updates or changes in economists' assumptions. [Excel]
- Built an R Shiny dashboard of comparable indicators for five countries, scoped with economists and linked in weekly division updates to support cross-country monitoring. [R, Shiny, APIs]
- Wrote cross-country COVID-19 policy memos for the mission chief briefing, highlighting fiscal burdens of contingency spending and SACU policy differences.

Global Risk Intelligence | Lead Data Scientist Jun 2019–Feb 2020
Washington, DC | Boutique global-risk consultancy

- Led three colleagues in designing and implementing a machine-learning project on aerospace supply-chain risk; presented findings and strategic recommendations to client stakeholders. [R, Tableau]
- Built a Python data pipeline using decision-tree classification of delivery delays to guide risk-based procurement diversification and clustering to identify comparable alternative suppliers. [Python]
- Developed a reputational-risk demo for a high-profile client using Twitter and public documents, Quanteda keyness testing, VADER sentiment analysis and structural topic modeling (STM). [R, Python, Excel]
- Delivered a Tableau risk dashboard and structured playbook documenting the analytical workflow for client handover; the client adopted the prototype across its entire risk division. [R, Tableau]

PUBLICATION

Thioune, F. K., Albertin, G., Kazandjian, R., & Wang, T. (2024). COVID-19 and Gender Inequality: Impact in Southern Africa. In *Gender Equality and Economic Development in Sub-Saharan Africa*, Chapter 4. International Monetary Fund.

SELECTED ANALYTICAL PROJECTS

IMF Lending Programs Evaluation and Digitalization <i>African Department lending module within the IMF modernization program</i> - Rebuilt a lending dataset of more than 50,000 rows covering 1952 onward, using SQL and Python to reconcile missing or inconsistent internal records against published IMF Staff Reports. - Built a Power BI monitoring module, presentation materials and weekly reporting for departmental use; surfaced delayed program assessments for front-office follow-up with country teams.	2021–2023
Using Sentiment to Measure Market Stress <i>IMF Big Data for Surveillance</i> - Constructed a volatility-weighted exchange-market-pressure chronology for 48 countries over 1990–2019 and compared crisis flags against three independently published benchmark chronologies. - Engineered lagged macroeconomic and supplied sentiment features for five countries; compared XGBoost, logistic regression, SVC and OneClassSVM across 80 runs using F1 and event-detection behavior.	2020–2021
Climate Policy Text Analysis <i>IMF Climate Innovation Challenge toolbox</i> - Collected Twitter posts using keyword-based API queries, analyzed sentiment and built a Python/Plotly dashboard to assess public reactions and potential acceptance barriers to new climate policies. - Supplied policy metadata (categories, countries and implementation years) and sentiment results for a module linking policy timelines with emissions breaks, and recommended design improvements.	2021–2023
Government R&D and Climate Technology Innovation <i>Patent data integration and interactive visualization</i> - Screened more than eight million patents to identify 576,706 climate-technology patents, linking patent records to firm characteristics and government-funding sources for research on climate innovation. - Constructed technology-spillover and product-market-rivalry measures using USPTO patents and Compustat data. - Built an R Shiny explorer for patent trends, influential patents, firms and federal agencies, with technology filters that allow users to explore trends and compare organizations within selected patent subclasses.	2024–2025

TEACHING EXPERIENCE

University of Georgia Teaching Assistant - AAEC 2580 Applied Microeconomic Principles, Spring 2024 (Greg Colson). - AAEC 4730 World Food Economics, Fall 2024 (Ellen McCullough). - AAEC 3690 Agribusiness Finance, Spring 2025 (Michael Adjemian).	2024–2025
Hong Kong Baptist University External Professional Assessed BUSI 4005 professional-track BBA projects at key milestones.	2021–present

TECHNICAL SKILLS

Research: Applied microeconomics, causal inference, panel econometrics, food prices and policy evaluation
Methods & tools: R, Python, SQL, Stata, Excel, ArcGIS, Power BI, Tableau and Shiny
Research computing: Large-scale data processing, high-performance computing (Linux/SLURM), and data validation